

# Link Analysis

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## 1 Introduction

This project aims to implement a ranking of authors based on the PageRank index. For this scope, the dataset used is the arXiv one published by Cornell university, which contains millions of scientific papers. The first 100000 papers were considered to make the testing faster, but the number can be changed easily by changing the variable MAX\_RECORDS.

## 2 Data organization

First, the module zipfile is used to extract all the data from the zip file. The raw data is cleaned by getting only information about the authors names and removing the authors that have less than two papers published. After this, to reduce memory usage, some structures are created. For example, paper\_counts stores the number of papers written by each author, eligible\_authors contains the authors that respect the minimum papers requirement, and edge\_weights, which store the co-authorship relationships.

## 3 Pre-processing techniques

The authors' names are parsed to get the same structure for every author, which is last name, first name. Using a for loop, the last name is extracted, removing spaces, and is put in the variable last. The same is done for the first name, put in the variable first. After this, each author name is analyzed and authors filtered so that those with fewer than MIN\_PAPERS papers are excluded from the analysis to decrease memory consumption. Then, for every paper, co-author pairs are generated and the more collaborations, the higher the edge weights.

## 4 Algorithm and implementation

The above steps are implemented in each paper, producing a weighted network. The edge weights are first initialized to zero through the defaultdict dictionary. Then , the eligible\_authors variable is transformed into a list in alphabetical

order. The variable `author_to_id` is created to store a dictionary inside it and give to each author a numeric ID, based on their ranking in the authors list. After reading the zip file, the algorithm keeps only the authors of the article taken into account. The author names are subsequently transformed into their corresponding IDs and are ordered. The next step is to produce all unique co-author pairs using the module `itertools`, increasing the weight of each pair of co-authors. A PageRank algorithm is implemented and it uses edge weights to get collaboration frequency. The damping factor, which represents the probability that a random surfer continues to click through links rather than going to a random page, is the standard one, 0.85.

## 5 Scalability

The dataset is not upload in its entirety, rather it's iterated row by row, accessing it as a sequential flow. Edge weights, kept in `defaultdict`, are millions. Since they are entirely held into the RAM, after the creation of the graph the dictionary is deleted to recover some space.

## 6 Description of experiments

To test the algorithm, the following variables are created: `MAX_RECORDS` to reduce the size of the dataset to the desired number of papers, `MIN_PAPERS` is equal to two so that authors with less than two papers are excluded from the analysis, `DAMPING` is 0.85 as told before, `TOP_N` is used to visualize the first 20 authors. The library `networkX` was used before and, while doing well when the number of maximum papers was set to one hundred thousand, it blew the RAM space when the number was set to a million, so it had to be changed with `iGraph` in order to be able to process millions of papers.

## 7 Results

The project is able to successfully create a PageRank index for the chosen dataset, transforming the large dataset in a graph representation. The top author is Zhang, Y, with more than one thousand papers. The number is extremely high, and this could be due to the fact that Zhang is in the top 5 most common surnames in China, with tens of millions of people having this surname.

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